INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

SPEEDOMETERS; TACHOMETERS; FLEXIBLE SHAFT ASSEMBLIES; FLEXIBLE SHAFT ASSEMBLY CORES, ADAPTERS, AND TRANSMITTERS

This specification is approved for use within the U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mechanical and electrical speedometers and tachometers, flexible shaft assemblies, flexible shaft assembly cores, adapters, and transmitters used with these instruments.

1.2 Classification. The speedometers, tachometers, shaft assemblies, cores, adapters, and transmitters shall be of the following types, groups, and classes as specified (see 6.2).

Type I - Mechanical
  Group A - Speedometer, fixed mounting.
  Group B - Tachometer, fixed mounting.
    Class 1 - Shaft assembly, flexible, metallic.
    Class 4 - Core, flexible shaft assembly.
    Class 5 - Adapter, speedometer, and tachometer, 90-degree drive.
    Class 6 - (see 6.3).
    Class 7 - Adapter, speedometer, and tachometer, straight drive.
    Class 8 - Adapter, coupling/bulkhead, speedometer and tachometer drive.
    Class 9 - Shaft assembly, flexible, plastic.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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Type II - Electrical
Group C - Speedometer, fixed mounting.
Group D - Tachometer, fixed mounting.
Class 2 - Transmitter, electrical, transmission or engine mounted.
Class 3 - Transmitter, electrical, distributer mounted.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-380	- Plastic Molding Material, Methacrylate.

MILITARY

MIL-W-80	- Window, Observation, Acrylic Base, Antielectrostatic, Transparent (for Indicating Instrument).
MIL-F-13927	- Fungus Resistance Test, Automotive Components.
MIL-C-22750	- Coating, Epoxy, VOC-Compliant.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurements of.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.

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MS35916	- Tachometer, Mechanical, Fixed Mounting (4,000 Rpm).
MS39021	- Speedometer, Mechanical (0-60 Mph).
MS39130	- Adapter, Speedometer-Tachometer, 90° Drive.
MS39132	- Adapter, Speedometer-Tachometer, Straight Drive.
MS51071	- Shaft Assembly, Flexible Metallic-Tachometer and Speedometer.
MS51072	- Core, Flexible Shaft Assembly, Tachometer and Speedometer.
MS52116	- Shaft Assembly: Flexible, Plastic, Tachometer, and Speedometer.
DS52140	- Speedometer Mechanical (0-100 Km/h).
MS53099	- Adapter, Coupling/Bulkhead, Speedometer-Tachometer Drive.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Navy Publication and Printing Services Office, Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

1236900	- Chemical Agent Resistant Coatings (CARC) Paint System Index.

(Copies of the drawings should be obtained from the contracting activity assembly or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A380	- Standard Practice for Use of the International System of Units (SI) the Modernization Metric System.
ASTM B117	- Salt Spray (Fog) Testing, Method of.
ASTM D3935	- Standard Specification for Polycarbonate.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Units furnished under this specification shall be products which are qualified for listing on the applicable qualified products list (QPL) at the time set for opening of bids (see 4.4 and 6.4).

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Materials shall be free from defects and imperfections that might adversely affect the serviceability or appearance of the finished product (see 4.7.1).

3.2.1 Dial. The dial shall be fabricated from metal (see 4.7.1).

3.2.2 Cover window. The cover window shall be fabricated from clear plastic conforming to MIL-W-80 or, if molded, conforming to type I, class 3 of L-P-380 or Group 1, Class 1 of ASTM D3935.

3.2.3 Recycled, virgin, and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.5.2).

3.3 Design and construction. The design and construction of the units shall be in accordance either with the applicable drawings or with the following applicable standards for the specified type, grade, and class (see 1.2 and 6.2): DS52140, MS35916, MS39021, MS39130, MS39132, MS51071, MS51072, MS52116, and MS53099. Conformance shall be verified in accordance with 4.7.1 or 4.7.2, as applicable.

3.3.1 Speedometer or tachometer.

3.3.1.1 Mounting. The instrument shall be provided with a bezel, and a means for mounting (see 4.7.2).

3.3.1.2 Connector. The speedometer or tachometer shall be provided with a connector for attachment of a flexible drive shaft assembly or transmitter cable, as applicable.

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3.3.1.3 Case.

3.3.1.3.1 Case window. The case window shall facilitate illumination of the dial from an outside source.

3.3.1.3.2 Dehumidifier. Silica gel desiccant, sufficient to dehumidify trapped air after the speedometer or tachometer is sealed, shall be confined in a container within the speedometer or tachometer case prior to assembly. The desiccant container shall be secured so that it will not become loose when the speedometer or tachometer is subjected to vibration (see 3.5.2).

3.3.1.4 Dial.

3.3.1.4.1 Marking. The face of the dial shall include the markings specified in the applicable MS/DS standards or drawings. The printing shall be commercial Gothic-style lettering of the size specified in table I. Unless otherwise specified in the applicable MS/DS standards or drawings, the dial face shall have a black background. Numerals, letters, and graduations shall be white conforming to color 37875 of FED-STD-595.

3.3.1.4.2 Fastening. The dial shall be secured to the body of the speedometer and tachometer in such a position that the marking on the face of the dial and the numerals on the wheels of the recording device shall be legible when illuminated through the case windows from an outside source.

TABLE I. Dial face marking.

Marking	Height (inches)	Height (mm)
Numerals for miles per hour scale	7/32	5.6
Numerals for kilometers per hour scale	5/32	4.0
Graduations, major, miles per hour scale	7/32	5.6
Graduations, major, kilometers per hour scale	1/8	3.2
Graduations, minor, miles per hour scale	1/8	3.2
"MILES PER HOUR" or "MPH"	1/8	3.2
"KILOMETERS PER HOUR" or "KM/H"	5/32	4.0
"REVOLUTIONS PER MINUTE" or "RPM"	1/8	3.2
"HOURS" and "10TH" or "100TH"	3/32	2.4
"MILES" and "10TH"	3/32	2.4
MS or Military part number	3/32	2.4
Manufacturer's identification	1/16 (max.)	1.6

3.3.1.5 Pointer. The pointer shall be attached to the indicator movement and shall be damped to reduce oscillation from vibration. The pointer shall not touch the dial or window. The shape of the pointer shall facilitate reading of any point in the range of the dial. At least the top half of the visible portion of the pointer shall be white conforming to color 37875 of FED-STD-595.

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3.3.1.5.1 Damping. The pointer shall be damped to prevent an oscillation of more than ± 1/2 miles per hour (mph) above 10 mph or ± 1 kilometer per hour (km/h) above 15 km/h on the speedometer dial, or ± 30 revolutions per minute (rpm) above 500 rpm on the tachometer dial when the speedometer or tachometer is operated directly off a drive shaft or transmitter, as applicable (see 3.4.1), having no fluctuation in speed.

3.3.1.6 Recording devices.

3.3.1.6.1 Odometer. The instrument, when furnished for use with a speedometer, shall be provided with a nonreset-type odometer, consisting of six numeral wheels located in the dial as shown in the applicable drawing or either MS39021 or DS52140. The odometer shall totalize and record up to 99,999.9 in multiples of 0.1 mile or 0.1 kilometer before automatically turning to zero.

3.3.1.6.2 Engine hour meter. The instrument, when furnished for use with a tachometer, shall be provided with a nonreset-type engine hour meter comprised of five or six numeral wheels located in the dial as shown in the applicable drawing or MS35916. The meter shall totalize and record up to 9,999.9 in multiples of 0.1 hour or up to 9,999.99 in multiples of 0.01 hour before automatically turning to zero.

3.3.1.6.3 Numeral wheel marking. Numerals on the speedometer, odometer, tachometer, and the engine hour meter shall be white on a black background, except that the decimal numerals shall be black on a white background. The height of the numerals shall be not less than 5/32 inch [4.0 millimeters (mm)].

3.3.2 Drive mechanism. Mechanical speedometers and tachometers shall be driven by a flexible shaft assembly, with adapters as required. Electrical meters shall be actuated by an electric impulse through an insulated cable.

3.3.2.1 Flexible shaft assembly. The flexible drive shaft assembly for mechanical instruments shall consist of a flexible outer casing either metallic (see MS51071) or plastic (see MS52116) and a flexible inner core. The outer casing shall be equipped with ferrules and nuts for attaching the assembly to both the instrument and the take-off point. The configuration shall fall within the envelope limits indicated in MS51071 or MS52116 or in the applicable drawing. End connections affecting interchangeability shall conform to detailed requirements. The attachment nuts shall be steel. Otherwise, construction may be of rubber, plastic or metal or any combination thereof. When plastic or rubber components are incorporated in the design, the assembly shall meet the ozone requirements specified in 3.5.7. Gaskets or washers shall be provided in the assembly to provide for waterproofness and to prevent the core from falling out.

3.3.2.1.1 Flexible shaft assembly, oscillation. When the flexible shaft assembly is tested as specified in 4.7.4.2.1, pointer oscillation shall not exceed: 1 mph at speeds above 10 mph [2 km/h at speeds above 15 km/h] on the speedometer dial; and 50 rpm on the tachometer dial at rotating speeds above 500 rpm.

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3.3.2.1.2 Core. The flexible shaft assembly core shall conform to the applicable drawing or MS51072. The outer layer of core wires shall be wound in the direction of right hand lay.

3.3.2.2 Adapters. The adapters shall conform to the applicable drawings or either MS39130, MS39132, or MS53099.

3.3.2.3 Transmitters. The electrical transmitters for electrical speedometers and tachometers shall be mountable either on the engine, transmission, or distributer as specified (see 1.2 and 6.2) and shall conform to the applicable drawings or standards.

3.4 Performance.

3.4.1 Calibration. When tested as specified in 4.7.3.1, calibration of the instruments shall be such as to meet the following performance requirements.

3.4.1.1 Speedometers (mechanical and electrical). When tested as specified in 4.7.3.1.1, the speedometer pointer shall indicate the vehicle speed specified in table II within the specified calibration limits when the drive shaft and transmitter is driven at the specified speeds after the speedometer temperature has been stabilized at ambient air temperatures of plus 77 ± 15 degrees Fahrenheit (°F) [25 ± 8 degrees Celsius (°C)], and plus 160 ± 5°F (71 ± 3°C). An additional 3 mph [5 km/h] tolerance above or below the recorded speed at 77°F (25°C) shall be allowed when the speedometer is tested a plus 160°F (71°C). The pointer shall leave the stop at or before the moment when the drive shaft and transmitter is being driven at a speed for which the correct indicated speed is 3 mph [5 km/h].

TABLE II. Mechanical and electrical speedometer calibration limits.

Drive shaft and transmitter speed (rpm)	Correct indicated speed		Calibration limits at 77 ± 15°F (25 ± 8°C)
	mph	km/h	
167	10	16	+1 to -1
500	30	48	+2 to -1
1000	60	96	+3 to -0

3.4.1.2 Odometer (performance). When tested as specified in 4.7.3.1.2, the odometer shall register within a tolerance of plus 1 percent (%).

3.4.1.3 Tachometers (mechanical and electrical). When tested as specified in 4.7.3.1.3, the tachometer pointer shall indicate a rotating speed on the dial twice that of the drive shaft or transmitter within a tolerance of ± 60 rpm. The drive shaft or transmitter shall be rotated at 250, 1000 and 1500 rpm. The test shall be conducted with the tachometer temperature stabilized at ambient air temperatures of 77 ± 15°F (25 ± 8°C) and plus 160 ± 5°F (71 ± 3°C). When tested as 160°F an

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flexible outer casing either

additional tolerance is required. These calibration tolerances are determined by multiplying the initial reading by 5% and adding and subtracting it to/from the initial tolerance upper and lower limits. The pointer shall leave the stop at or before the moment when the drive shaft is being driven at a speed for which the correct indicated speed is 100 rpm.

3.4.1.4 Engine hour meter (performance). When tested as specified in 4.7.3.1.4, the total limit of engine operation during each phase of testing shall be recorded in hours by the engine hour meter on the tachometer within ± 2%.

3.4.1.5 Transmitters. When tested as specified in 4.7.3.1.1 through 4.7.3.1.4, with an approved (or previously calibrated) electrical speedometer and tachometer, the transmitter shall cause the speedometer and tachometer to operate within the applicable calibration limits.

3.4.1.6 Flexible shaft and core assemblies (performance). When tested as specified in 4.7.3.1.5, the flexible shaft and core assemblies shall not exceed ± 2 mph (± 3 km/h) speedometer pointer oscillation.

3.4.2 Resistance to wear.

3.4.2.1 Speedometer, tachometer, and transmitter. After testing as specified in 4.7.4.1, the speedometers, tachometers, and transmitters shall meet the previously recorded readings obtained under 3.4.1 within ± 3%.

3.4.2.2 Flexible shaft assembly. When tested as specified in 4.7.4.2, the flexible shaft assembly shall withstand operation for 500 hours without failure.

3.4.2.3 Adapters (resistance). When tested as specified in 4.7.4.2.2, adapters of each class shall withstand operation for 500 hours without failure.

3.4.3 Flexible shaft assembly and adapters.

3.4.3.1 Operating torque (flexible shaft assembly). When tested as specified in 4.7.5.1 at an ambient air temperature of minus (-) 65 ± 5°F (-54 ± 3°C), the torque required to rotate the flexible drive shaft core (see 4.3.2) while starting and uniformly accelerating from 0 to 1000 rpm in 15 seconds shall at no time exceed 80 ounce-inches [0.56496 Newton meter (N.m)].

3.4.3.2 Flexibility (flexible shaft assembly, plastic only). When tested as specified in 4.7.5.2, the casing of the flexible shaft assembly shall be in contact with the mandrel over the top 180 degrees (°) of the mandrel circumference with the load applied at the free end not exceeding an 8-pound [3.6-kilogram (kg)] pull.

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3.4.3.3 Adapters. When tested as specified in 4.7.5.3 and at an ambient air temperature of -65 ± 5°F (-54 ± 3°C), the torque required to rotate adapters of each class while uniformly accelerating from 0 to 1000 rpm in 15 seconds shall at no time exceed 80 ounce-inches (0.56496 N.m).

3.4.3.4 Torsional strength (flexible shaft assembly). When tested as specified in 4.7.5.4, the flexible shaft assembly core shall have a torsional breaking strength of not less than 15 pound-inches (0.1695 N.m).

3.4.3.4.1 Pull strength, end fittings. When tested as specified in 4.7.5.4, end fittings attached to the core shall withstand a pull of 50 pounds (22.7 kg).

3.5 Environmental.

3.5.1 Lubrication leakage (lubricated flexible shaft assembly only). The flexible drive shaft assembly shall withstand the test specified in 4.7.6.1 without evidence of leakage of lubricant. Subsequent to the test, the drive shaft assembly shall meet the requirements of 3.4.3.1.

3.5.2 Vibration resistance. When tested in accordance with 4.7.6.2, the instrument and transmitter shall withstand vibration for 1 hour in each direction of the three major axes without evidence of disintegration of component materials or assemblies. At the conclusion of the test, the instrument and transmitters shall meet the previously recorded readings obtained under 3.4.1 within ± 3%.

3.5.3 Shock resistance. When tested in accordance with 4.7.6.3, speedometers, tachometers, and transmitters shall show no evidence of sticking of the pointer, no evidence of delay of functioning, and no evidence of disintegration of component materials or assemblies. At the conclusion of the test, instruments and transmitters shall meet the previously recorded reading obtained under 3.4.1 within ± 3%.

3.5.4 Waterproofness.

3.5.4.1 Waterproofness, instruments and transmitter. When tested in accordance with 4.7.6.6, instruments and transmitters shall meet the waterproofness requirements specified for type I, class 2 components in MIL-STD-1184. When tested as specified in 4.7.6.7.1, speedometers, tachometers, and transmitters shall show no evidence of leakage.

3.5.4.2 Waterproofness, flexible shaft assemblies and adaptors. When tested as specified in 4.7.6.7 and 4.7.6.7.1, the flexible shaft assemblies and adapters shall show no sign of leakage.

3.5.5 Corrosion resistance. When tested in accordance with 4.7.6.5, speedometers, tachometers, flexible shaft assemblies, transmitters, and adapters shall show no evidence of corrosion affecting operation and shall subsequently meet the applicable requirements of 3.4.1.

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3.5.6 Fungus resistance. When tested as specified in 4.7.6.6, speedometers, tachometers, flexible shaft assemblies, transmitters and adapters shall show no evidence of fungus growth affecting operation and shall subsequently meet the applicable requirements of 3.4.1.

3.5.7 Ozone resistance. When tested as specified in 4.7.6.7, the flexible shaft assembly shall show no evidence of cracking, rupture, or other deterioration.

3.5.8 Electromagnetic interference (electrical speedometers, tachometers, and transmitters only). When tested as specified in 4.7.6.7.1, electrical speedometers, tachometers, and transmitters shall meet the electromagnetic interference requirements for class IIC equipment of MIL-STD-461.

3.6 Painting. Unless otherwise specified, cleaning, pretreatment, priming and painting shall conform to the appropriate CARC paint system selected from Drawing No. 12369000. When instrument bezels made of stainless steel are used, they shall be passivated per ASTM A380, primer washed per DOD-P-15328, and finish painted per MIL-C-46168, color Green 383, to a dry film thickness of 1.8 to 3.2 mils. When mechanical speedometers and tachometers (MS39021 and MS35916, respectively) are used in the interior of a vehicle or system and epoxy finish topcoat per MIL-C-22750, color lusterless Green 34094 or 34095, to a dry film thickness of 1.3 to 1.7 mils shall be used.

3.7 Identification and marking. Identification and marking of the items shall be in accordance with MIL-STD-130. As a minimum it shall consist of the manufacturer's CAGE code and the applicable military part number. All markings shall be permanently and legibly printed or stamped on the dial of the speedometers and tachometers, on the casing of the flexible shaft assemblies, and on the housing of the adapters and the transmitters. The drive ratio shall be marked on the housing of all adapters. The flexible shaft assembly cores shall be tagged (see 4.7.2 and 6.2).

3.8 Workmanship. Workmanship shall be such as to assure a product free of rust, burrs, and scratches. All electrical connections shall be mechanically secured before being soldered or brazed, to preclude the possibility of becoming loose due to vibration or shock. All solder cleaning agents and flux shall be noncorrosive. All trace of flux shall be chemically neutralized and removed from soldering (see 4.7.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize the submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.4).
b. Quality conformance inspection (QCI) (see 4.5).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions.

4.3.1 Conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government. Unless otherwise specified herein, all tests shall be conducted at ambient room air temperature. The ambient room temperature shall be maintained at 77 ± 15°F (25 ± 8°C). Unless otherwise specified herein or in applicable specifications, test specimens shall be thermally stabilized for 1 hour prior to being subjected to tests.

4.3.2 Standard specimens. Unless otherwise specified herein and excepting acceptance tests, the shaft assembly shall conform to MS51071-5 or MS52116-5.

4.3.3 Apparatus. In addition to the standard measuring equipment, test apparatus shall include a temperature-controlled air chamber, a submersion tank, and a test stand with suitable drives for rotating the drive shaft or transmitter at controlled speeds.

4.4 Qualification inspection. Qualification inspection shall consist of all the examinations and tests specified under 4.7 and as listed in table III. The qualification sample shall be representative of the items proposed to be supplied under contract. Qualification inspection shall be conducted under Government surveillance by the contractor or by a testing facility approved by the Government.

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TABLE III. Classification of inspections.

Title	Requirement	Inspection	Qualification	QCI		Control
				Examination	Tests	
Group A:						
Materials and construction	3.2 through 3.3.2.3, 3.6	4.7.1	X			
Defects	3.3, 3.6, 3.7, 3.8	4.7.2	X	X		
Group B (performance):						
-Calibration						
Speedometer	3.4.1.1	4.7.3.1.1	X			X
Odometer	3.4.1.2	4.7.3.1.2	X			X
Tachometer	3.4.1.3	4.7.3.1.3	X			X
Engine-Hour meter	3.4.1.4	4.7.3.1.4	X			X
-Wear						
Speedometers, tachometers, and transmitters	3.4.2.1	4.7.4.1	X			
Flexible shaft assy. & adapter	3.4.2.2	4.7.4.2	X			
Flexible shaft assembly oscillation	3.3.2.1.1	4.7.4.2.1	X			
Adapters	3.4.2.3	4.7.4.2.2	X			
-Torque						
Flexible shaft assembly	3.4.3.1	4.7.5.1	X			
Flexibility (plastic only)	3.4.3.2	4.7.6.2	X			
Adapters	3.4.3.3	4.7.5.2	X		X	
Torsional strength	3.4.3.4	4.7.5.3	X			
Pull strength	3.4.3.4.1	4.7.5.3	X			
Group C (environmental):						
Lubricant-leakage (flexible shaft assembly only)	3.5.1	4.7.6.1	X			
Vibration	3.5.2	4.7.6.3	X			
Shock	3.5.3	4.7.6.4	X			
Waterproofness:						
Instruments and transmitters	3.5.4.1	4.7.6.6	X			
		4.7.6.7.1			X	
Flexible shaft assemblies and adapters	3.5.4.2	4.7.6.7	X			
		4.7.6.7.1			X	

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TABLE III. Classification of inspections - Continued.

Title	Requirement	Inspection	Qualification	QCI		Control
				Examination	Tests	
Corrosion resistance	3.5.5	4.7.6.5	X			
Fungus resistance	3.5.6	4.7.6.6	X			
Ozone resistance	3.5.7	4.7.6.7	X			
Electromagnetic interference	3.5.8	4.7.6.8	X			

4.4.1 Qualification sample. The qualification sample and sample saize shall consist of the items listed in table IV (see 6.2).

TABLE IV. Qualification sample and sample size.

Sample item	Sample size
Type I, group A speedometer, mechanical	3
Type II, group C speedometer, electrical	3
Type I, group B tachometer, mechanical	3
Type II, group D tachometer, electrical	3
Type I, class 1 or class 9 shaft assembly, flexible; mechanical speedometer and tachometer	4
Type I, class 5 adapter, speedometer and tachometer 90 degree drive	4
Type I, class 7 adapter, speedometer and tachometer straight drive	4
Type I, class 8 adapter, coupling/bulkhead-speedometer and tachometer drive	4
Type II, class 2 transmitter, electrical, transmission or engine mounted	3
Type II, class 3 transmitter, electrical, distributor mounted	3

4.4.2 Qualification test sequence. The qualification sample selected from table IV for qualification inspection shall be tested in accordance with the test sequence specified in table V for the applicable item.

TABLE V-A. Sequence of qualification testing for speedometers and tachometers.

Speedometers and tachometers	
Specimen A:	
4.7.4.1	Wear
4.7.6.4.8	Waterproofness
4.7.3.1.1	1/

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TABLE V-A. Sequence of qualification testing for speedometers and tachometers - Continued.

Specimen B:	
4.7.3.1.1	1/
4.7.6.3	Shock
4.7.6.6	Fungus resistance
Specimen C:	
4.7.3.1.1	1/
4.7.6.2	Vibration
4.7.6.5	Corrosion resistance

1/ = Test at ambient temperature (77 ± 15°F/25 ± 8°C).

TABLE V-B. Sequence of qualification testing for flexible shaft assemblies.

Flexible shaft assemblies	
Specimen A:	
4.7.6.4.3	Waterproofness
4.7.4.2	Wear
4.7.6.5	Corrosion resistance
Specimen B:	
4.7.6.1	Lubrication leakage
4.7.5.1	Operating torque
Specimen C:	
4.7.5.2	Flexibility (plastic only)
4.7.4.2.1	Flexibility shaft assembly, oscillation
4.7.5.7	Fungus resistance
Specimen D:	
4.7.5.4	Torsional strength
4.7.6.7	Ozone resistance

TABLE V-C. Sequence of qualification testing for adapters.

Adapters	
Specimen A:	
4.7.4.2.2	Wear
4.7.6.4.3	Waterproofness
Specimen B:	
4.7.6.5	Corrosion resistance

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TABLE V-C. Sequence of qualification testing for adapters - Continued.

Specimen C:	
4.7.6.6	Fungus resistance
Specimen D:	
4.7.5.2	Operating torque

TABLE V-D. Sequence of qualification testing for transmitters.

Transmitters	
Specimen A:	
4.7.3.1.1	1/
4.7.6.3	Vibration
4.7.4.1	Wear
Specimen B:	
4.7.3.1.1	1/
4.7.6.3	Shock resistance
4.7.6.6	Fungus resistance
Specimen C:	
4.7.3.1.1	1/
4.7.6.6	Waterproofness
4.7.6.5	Corrosion resistance

4.4.3 Qualification failure. Deficiencies/faults found during, or as a result of, qualification inspection may be cause for the Government to either discontinue qualification testing or to refuse to conduct additional testing until the deficiencies/faults have been corrected. The Government will not approve products for listing on the qualified products list (QPL) until qualification inspection is completed to the satisfaction of the Government.

4.4.4 Retention of qualification. Certification shall be requested every 2 years from each manufacturer listed on the QPL to retain listing on the QPL. This certification shall be forwarded to the preparing activity and shall be signed by a responsible official of management, attesting that the listed product still meets the requirements of the current issue of the specification, is available from the listed plant, and can be produced under the same conditions as originally qualified; i.e., same process, materials, construction, design, manufacturers part number or designation. Failure to provide certification shall be cause for removal from the QPL.

4.5 QCI. QCI shall include the examination of 4.5.2 and the test of 4.5.3. Noncompliance with any of the specified requirements in sections 3 and 5 shall be cause for rejection of the sample and the inspection lot.

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4.5.1 Sampling plan. Unless otherwise specified (see 6.2), the sampling plan specified herein shall be used. See 6.4 for definitions of sampling inspection terms.

4.5.1.1 Lot formation. An inspection lot shall consist of all speedometers, tachometers, flexible shaft assemblies of a single type, class, style and part identification number (PIN), from an identifiable production period, from one manufacturer, from one manufacturing location, submitted at the same time for acceptance.

4.5.1.2 Sample. The sample for QCI examination and tests shall be randomly selected from the inspection lot in accordance with table VI.

TABLE VI. Sampling plan for QCI.

QCI sampling plan			
Inspection lot size	Sample size		
	Examination		Test
	Major	Minor	
2 to 8	*	5	5
9 to 15	13	5	5
16 to 25	13	5	5
26 to 50	13	5	5
51 to 90	13	7	7
91 to 150	13	11	11
151 to 280	20	13	13
281 to 500	29	16	16
501 to 1200	34	19	19
1201 to 3200	42	23	23
3,201 to 10,000	50	29	29
10,001 to 35,000	60	35	35
35,001 to 150,000	74	40	40
150,001 to 500,000	90	40	40
500,001 and over	102	40	40

* Indicates entire lot must be inspected (100% inspection).

4.5.2 Examination. The sample selected in accordance with 4.5.1.2 shall be examined and defects classified as specified in tables V (see 4.7.2). The acceptance number in all cases is zero.

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TABLE VII. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major		
101	Incorrect dimensions affecting interchangeability (see 3.3).	SIE 1/
102	Faulty workmanship, function (see 3.8).	Visual
Minor		
201	Incorrect dimensions not affecting interchangeability (see 3.3).	SIE
202	Improper painting (see 3.6).	Visual
203	Improper identification and marking (see 3.7).	Visual
204	Faulty workmanship, appearance (see 3.8).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Tests. The sample selected in accordance with 4.5.1.2 shall be subjected to the tests specified in table III. The acceptance number in all cases is zero.

4.5.4 QCI failure. Any item that fails to conform to any specified requirement shall be rejected; any failure (one or more) of the selected sample in either the Major/Minor categories or test for the appropriate inspection lot size shall constitute a failure of the entire lot. The rejected item(s) may be repaired or corrected and resubmitted for inspection. If the contractor utilizes sampling inspection as an element of his inspection system, rejected inspection lots may be resubmitted for acceptance if the contractor performs 100 percent inspection on the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units or obtains procuring activity approval to resample the lot due to the insignificance of the defects. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots.

TABLE VIII. Quality conformance tests.

Drive shafts and adapters	
4.7.6.5.3	Waterproofness
4.7.5.3	Torsional strength (drive shaft only)

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TABLE IX. Control tests.

Speedometers, tachometers and transmitters	
4.7.3.1.1	Speedometer calibration
4.7.3.1.2	Odometer calibration
	or
4.7.3.1.3	Tachometer calibration
4.7.3.1.4	Engine hour meter calibration

4.6 Control tests. Control tests shall be conducted on three units from each lot of 1000 units consecutively produced, except that not more than six tests shall be performed in a one month period, nor less than three tests in a one month period. The unit shall be subjected to the control tests specified in table III in the order listed in table IX.

4.6.1 Failure. If the units selected fail to pass any of the control tests, the Government inspector shall stop acceptance of subsequent items until the faults revealed have been corrected. Any defect found during or as a result of the test shall be prima facie evidence that items accepted after the previous acceptable control test are similarly defective. Such defects on all items shall be corrected by the contractor at no cost to the Government.

4.7 Methods of inspection.

4.7.1 Materials, design and construction. Conformance to 3.2 thru 3.3.2.3, and 3.6 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.3, 3.6, 3.7, and 3.8 shall be determined by examination for defects listed in table VII. Examination shall be visual, tactile, or by measurement with standard inspection equipment (SIE).

4.7.3 Performance.

4.7.3.1 Calibration.

4.7.3.1.1 Mechanical and electrical speedometers. To determine conformance to 3.4.1.1, the speedometers and tachometers while connected to a flexible shaft assembly or transmitter, as applicable, shall be mounted in an air chamber with the dial face tilted 10° backward from the vertical. The air chamber shall be maintained at each of the following temperatures for 4 hours: 77 ± 15°F (25 ± 18°C) and 160 ± 5°F (71 ± 3°C). After each

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4-hour period and while still maintaining the temperature, the drive shaft and transmitter shall be rotated at 167, 500, and 1000 rpm. The electrical speedometer shall be energized at 28.5 volts (V) and shall be operated for 5 minutes before recording the reading.

4.7.3.1.2 Odometer. To determine conformance to 3.4.1.2, at the conclusion of the test specified in 4.7.3.1.1, rotation of the drive shaft and transmitter shall be continued for 6 hours at a speed of 1000 rpm in an ambient air temperature of 77 ± 15°F (25 ± 8°C). The odometer indicated mileage or kilometers for the 6-hour period shall be recorded and compared with the correct mileage or kilometers.

4.7.3.1.3 Tachometers (mechanical and electrical). To determine conformance to 3.4.1.3, tachometers shall be tested as specified in 4.7.3.1.1, except that the drive shaft and transmitter shall be rotated at 250, 1000, and 1500 rpm. Electrical tachometers shall be energized at 28.5 V and operated for 5 minutes before recording the indicator readings.

4.7.3.1.4 Engine hour meter. To determine conformance to 3.4.1.4, at the conclusion of the test specified in 4.7.3.1.3, the drive shaft and transmitter shall be rotated at one of the optional speeds and for the time length specified in table IX. During and after the operating period of testing, the engine hour meter reading shall be observed.

TABLE X. Tachometer hour meter optional test speeds.

Tachometer type	Rpm	Time
I	1000	6 hours, 9.6 minutes
II	1000	6 hours, 36 minutes
I	1026	6 hours
II	1100	6 hours

4.7.3.1.5 Flexible shaft and core assemblies. To determine conformance to 3.4.1.6, the flexible shaft and core assemblies shall be connected to a previously calibrated speedometer and shall be rotated to indicate 35 mph (56 km/h) for a period of 10 minutes. The speedometer shall be observed for any deviations from the requirements of 3.4.1.6.

4.7.4 Wear test.

4.7.4.1 Speedometers, tachometers, and transmitters. To determine conformance to 3.4.2.1, the speedometers, tachometers, and transmitters shall be operated for 1000 hours at the speeds and temperatures specified in table XI. At the conclusion of the 1000-hour test period, the speedometers, tachometers, and transmitters shall be subjected to the applicable calibration test, at 77 ± 15°F (25 ± 8°C), specified in 4.7.3.1.

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TABLE XI. Speedometer and tachometer wear test.

Hours	Temperature	Speed
750	77 ± 15°F (25 ± 8°C)	Equal indicating period for 1/3, 2/3, and full scale reading (same as above)
250	160 ± 5°F (71 ± 3°C)	

4.7.4.2 Flexible shaft assembly. To determine conformance to 3.4.2.2, a flexible shaft assembly 10 feet (3.05 meters) long shall be mounted in a test fixture in such a manner that the shaft is confined in a S-shape having two bends of 180° in the same plane with each bend having a 5-inch (12.7-cm) radius. With a constant load of 22 ± 2 ounce-force-inches (0.155 ± 0.0141 N.m) at the speedometer and tachometer end of the shaft core, the core shall be operated at the speeds, temperatures, and times specified in table XII.

TABLE XII. Flexible shaft assembly and adapter wear test.

Hours	Temperature		Speed (rpm)
	°F	°C	
100	77 ± 15	25 ± 8	750
100	77 ± 15	25 ± 8	1250
200	77 ± 15	25 ± 8	2000
25	160 ± 5	71 ± 3	750
25	160 ± 5	71 ± 3	1250
50	160 ± 5	71 ± 3	2000

4.7.4.2.1 Flexible shaft assembly, oscillation. To determine conformance to 3.3.2.1.1, a 10-foot (3.05-meter) length of flexible shaft assembly shall be connected to a previously calibrated speedometer and shall be rotated at indicated speedometer speeds from 10 to 60 mph at 10 mph intervals or 20 to 100 km/h at 20 km/h intervals. Running time at each speed shall then be repeated with a previously calibrated tachometer. The indicated tachometer rpm shall be between 500 and 4000 at 500 rpm intervals with a running time of 10 minutes at each rotating speed.

4.7.4.2.2 Adapters. To determine conformance to 3.4.2.3, the adapter shall be mounted in a fixture with a constant load of 22 ± 2 ounce-force-inches (0.155 ± 0.0141 N.m) and operated at the speeds, temperatures, and times specified in table XII.

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4.7.5 Operating torque.

4.7.5.1 Flexible shaft assembly. To determine conformance to 3.4.3.1, the flexible shaft assembly (see 4.3.2) shall be placed in a test fixture with the test sample temperature and air temperature maintained constant at -65 ± 5°F (-54 ± 3°C) for 7 hours. At the conclusion of the 7-hour period and while still held at the above specified temperature, the core shall be rotated without load. The torque required to rotate the core in starting and accelerating shall be measured.

4.7.5.2 Flexibility (plastic only). To determine conformance to 3.4.3.2, a section of flexible shaft assembly, approximately 17 inches (43 cm) from one end, shall be bent 180° around a mandrel having a diameter of 10 inches (25.4 cm) in an ambient air temperature of 77 ± 15°F (25 ± 8°C). A spring scale or other suitable measuring device shall be used to record the maximum pull required, at the fitting end.

4.7.5.3 Adapters (operating torque). To determine conformance to 3.4.3.3, the adapter shall be placed in a test fixture with the test sample temperature and air temperature maintained constant at -65 ± 5°F (-54 ± 3°C) for 7 hours. At the conclusion of the 7-hour period and while still held at the above specified temperature, the adapter shall be rotated without load. The torque required to rotate the adapter in starting and accelerating shall be measured.

4.7.5.4 Torsional strength (flexible shaft assembly). To determine conformance to 3.4.3.4, the flexible shaft assembly core shall be placed in a torsion-testing machine in a straight position without slack so that 10 inches (25.4 cm) of the speedometer and tachometer end of the core will be subjected to torsion. The core shall be twisted in a clockwise direction. The test shall be repeated with the drive end of the core in the machine. The core end fittings shall be included in the 10-inch (25.4 cm) length tested. The core end fittings shall then be subjected to a pull test with the fittings anchored and the load applied to the core to determine conformance to 3.4.3.4.1.

4.7.6 Environmental.

4.7.6.1 Lubricant-leakage (flexible shaft assembly only). To determine conformance to 3.5.1, the flexible shaft assembly shall be carefully sealed at both ends and placed in a temperature-controlled air chamber at a temperature of 250 ± 5°F (121 ± 3°C), and held at that temperature for 2 hours. At the conclusion of the 2-hour period, the assembly shall be removed from the oven and suspended from the transmission end in a vertical position. The flexible shaft assembly shall be observed for evidence of leakage of lubricant. Subsequently, the flexible shaft assembly shall be subjected to the test specified in 4.7.5.1 to determine conformance to 3.4.3.1.

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4.7.6.2 Vibration. To determine conformance to 3.5.2, the speedometer, tachometer, and transmitter shall be mounted as if in operation and subjected to the vibration test as specified in MIL-STD-202, method 201, except that motion shall be applied for 1 hour in each direction of the three major axes. During vibration, the speedometers and tachometers shall be operated in such a manner that the speedometer and tachometer pointer shall be at midscale, and observation shall be made for oscillation. At the conclusion of the test, the speedometer, tachometer, and transmitter shall be subjected to the applicable 77 ± 15°F (25 ± 8°C) calibration test specified in 4.7.3.1.

4.7.6.3 Shock. To determine conformance to 3.5.3, the speedometer, tachometer, and transmitter shall be mounted as if in operation and subjected to the shock test as specified in MIL-STD-202, method 213, test condition I (using the sawtooth waveform). The test shall be repeated 10 times and applied in the vertical axis. During shock, the speedometer and tachometer shall be operated in such a manner that the speedometer and tachometer pointer shall be at midscale and observation shall be made for oscillation. At the conclusion of the test, the speedometer, tachometer, and transmitter shall be subjected to the applicable 77 ± 15°F (25 ± 8°C) calibration test specified in 4.7.3.1.

4.7.6.4 Waterproofness.

4.7.6.4.1 Waterproofness, qualification test for instruments and transmitters. To determine conformance to 3.5.4.1, the speedometers, tachometers, and transmitters shall be tested for waterproofness as specified in MIL-STD-1184, method 100, procedure I, steps 1, 2, and 3. The speedometer and transmitter connectors shall be sealed with caps simulating the mating flexible cable connectors.

4.7.6.4.2 Waterproofness, qualification test for flexible shaft assemblies and adapters. To determine conformance to 3.5.4.2, the flexible shaft assemblies and adapters shall be subjected to an internal pressure of 6 pounds per square inch (psi) (0.42 kilograms per square centimeter) for 30 minutes while totally immersed in tap water. Observation for air bubbles from the interior of the component shall be maintained during the 30-minute period. Bubbles which are the result of entrapped air on the various exterior surfaces of the component shall not be considered a leak. The flexible shaft assemblies and adapters shall be sealed against leakage.

4.7.6.4.3 Waterproofness, quality conformance test. To determine conformance to 3.5.4.1 and 3.5.4.2, the speedometers, tachometers, flexible shaft assemblies, transmitters, and adapters shall be subjected to an internal pressure of 6 psi (0.42 kilograms per square centimeter) for 5 minutes while totally immersed in tap water. Observation for air bubbles from the interior of the component shall be maintained during the 5-minute period. Bubbles which are the result of entrapped air on the various exterior surfaces of the component shall not be considered a leak. The speedometer and transmitter connectors and both ends of the flexible shaft assemblies and adapters shall be sealed against leakage with connectors simulating the applicable mating parts or as otherwise specified.

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4.7.6.5 Corrosion resistance. To determine conformance to 3.5.5, the speedometers, tachometers, flexible shaft assemblies, transmitters, and adapters shall be subjected to the corrosion resistance test as specified in ASTM B117 for 200 hours. At the conclusion of the 200-hour period, the speedometers, tachometers, flexible shaft assemblies, transmitters, and adapters shall be subjected to the applicable test specified in 4.7.3.1. The speedometer and transmitter connectors and both ends of the flexible shaft assemblies and adapters shall be sealed against leakage with connectors simulating the applicable mating parts or as otherwise specified.

4.7.6.6 Fungus resistance. To determine conformance to 3.5.6, the speedometers, tachometers, flexible shaft assemblies, transmitters, and adapters shall be subjected to the fungus resistance test as specified in MIL-F-13927, method A, class 1 or 3. After the test, the speedometers, tachometers, flexible shaft assemblies, transmitters, and adapters shall be subjected to the applicable test specified in 4.7.3.1. The speedometer and transmitter connectors and both ends of the flexible shaft assemblies and adapters shall be sealed against leakage with connectors simulating the applicable mating parts or as otherwise specified.

4.7.6.7 Ozone resistance. To determine conformance to 3.5.7, the test procedure specified in 4.7.6.7 shall be followed using the apparatus specified in 4.3.3 and 4.7.6.6.

4.7.6.7.1 Apparatus. The apparatus shall include an exposure chamber containing air mixed with ozone in the proportion of 50 ± 5 parts per 100,000,000 parts of atmospheric air by volume.

4.7.6.7.2 Procedure. The specimen shall be conditioned at ambient room temperature for 45 minutes. The specimen shall then be placed in the test exposure chamber for 168 hours, at a temperature of 100 ± 5°F (38 ± 3°C). Immediately after the completion of the test period, each sample shall be examined for evidence of cracking using 7X magnification. The flexible shaft assembly specimen shall be coiled in a 12-inch (30-cm) inside-diameter loop.

4.7.6.8 Electromagnetic interference. To determine conformance to 3.5.8, the instruments and transmitters shall be tested in accordance with test methods CE03 and RE02 of MIL-STD-462.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. Speedometers, tachometers, and related parts covered by this specification are intended for use in military motor vehicles to indicate vehicle speed in miles per hour or kilometers per hour, or engine speed in revolutions per minute, and to record total distance in miles or kilometers traveled by the vehicle or total hours of operation of the engine.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, group, and class required and applicable drawing or standard part number (see 1.2, 3.3 and 3.7).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. For type I units, the applicable MS standard and for type II units, the applicable drawing (see 1.2 and 3.3).
e. Military part number or MS/DS standard (see 3.7).
f. Whether type II, group D, class 2 electrical transmitters should be engine or transmission mounted (see 1.2 and 3.3.2.3).
g. If painting is other than as specified (see 3.6).
h. If responsibility for inspection is other than as specified (see 4.1).
i. If inspection conditions are other than as specified (see 4.3).
j. Qualification inspection category (categories) (see 4.4 and 6.4).
k. Sample size for each qualification inspection category (see table IV).
l. Sample size for QCI examination and acceptance tests (see 4.5.1 and 4.5.2), acceptance criteria and disposition of defective items.
m. Selection of applicable level and packaging requirements (see 5.1).

6.3 Cancelled classification. Class 6, adapter, tachometer drive, 90° right and left rotation, MS39131, has been cancelled and replaced by MS39130.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time of award of contract, qualified for inclusion in the Quality Products List, whether or not such products have actually been so listed by that date. The attention of contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commanding General, U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, and information pertaining to qualification of products may be obtained from that activity.

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6.5 Definitions.

6.5.1 Definitions of terms used in sampling inspection.

a. Classification of defects. A classification of defects is the enumeration of possible defects of the unit of product classified according to their seriousness. A defect is any nonconformance of the unit of product with specified requirements. Defects will normally be grouped into one or more of the following classes: critical, major and minor defects. Also, defects may be grouped into other classes, or into subclasses within these classes.

b. Critical defects. A critical defect is a defect that judgement and experience indicate would result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product, or a defect that judgement and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

c. Critical defective. A critical defective is a unit of product which contains one or more critical defects and may also contain major and/or minor defects.

d. Defective. A defective is a unit of product which contains one or more defects.

e. Formation of lots or batches. The product shall be assembled into identifiable lots, sublots, batches, or in such other manner as may be prescribed (see 1). Each lot or batch shall, as far as is practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time.

f. Lot or batch. The term lot or batch shall mean "inspection lot" or "inspection batch", i.e., a collection of units or product from which a sample is to be drawn and inspected and may differ from a collection of units designated as a lot or batch for other purposes (e.g., production, shipment, etc.).

g. Lot or batch size. The lot or batch size is the number of units of product in a lot or batch.

h. Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose.

i. Major defective. A major defective is a unit of product which contains one or more major defects, and may also contain minor defects but contains no critical defect.

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j. Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

k. Minor defective. A minor defective is a unit of product which contains one or more minor defects but contains no critical or major defect.

l. Presentation of lots or batches. The formation of the lots or batches, lot or batch size, and the manner in which each lot or batch is to be presented and identified by the supplier shall be designated or approved by the responsible authority. As necessary, the supplier shall provide adequate and suitable storage space for each lot or batch, equipment needed for proper identification and presentation, and personnel for all handling of product required for drawing of samples.

m. Representative sampling. When appropriate, the number of units in the sample shall be selected in proportion to the size of sublots or subbatches, or parts of the lot or batch, identified by some rational criterion. When representative sampling is used, the units from each part of the lot or batch shall be selected at random.

n. Sample A sample consists of one or more units of product drawn from a lot or batch, the units of the sample being selected at random without regard to their quality. The number of units or product in the sample is the sample size.

o. Sampling plan. A sampling plan indicates the number of units of product from each lot or batch which are to be inspected (sample size or series of sample sizes) and the criteria for determining the acceptability of the lot or batch (acceptance and rejection numbers).

p. Time of sampling. Samples may be drawn after all the units comprising the lot or batch have been assembled, or samples may be drawn during assembly of the lot or batch.

6.5.2 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.5.3).

6.5.3 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

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6.6 Part or Identifying Number (PIN). Except when part numbers are specified in the drawing, MS standard or military specification sheet (MS sheet), the PIN to be used for the items acquired to this specification should be developed as follows:

M	10215	-	X	-	X	-	X
Prefix to indicate military specification	Specification number		Type number (1 = Type I, 2 = Type II) (see 1.2)		Group letter (see 1.2)		Class number (see 1.2)

6.7 Subject term (key word) listing.

Electrical connectors, vehicle
Electrical gages, transmission and engine
Gages, vehicle, electrical and mechanical
Instrument panel gages and connectors
Instrumentation, vehicle
Mechanical gages, transmission and engine
Recording devices, vehicle

6.8 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Material Command (AMC) policy for the elimination of AQLs/LTPDs from military specifications.

6.9 SI values. The English units are the primary units and the SI values are approximately equivalent to the primary units.

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Review activities:
Army - MI
DLA - GS

User activities:
Army - ME
Navy - MC

Preparing activity:
Army - AT

(Project 6680-0230)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-10215H	2. DOCUMENT DATE (YYMMDD) 921104

3. DOCUMENT TITLE Speedometers; Tachometers; Flexible Shaft Assemblies; Flexible Shaft Assembly Cores, Adapters, and Transmitters

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(313) 574-8774	786-8774
c. ADDRESS (Include Zip Code) COMMANDER U.S. ARMY TANK - AUTOMOTIVE COMMAND ATTN: AMSTA-GDS WARREN, MICHIGAN 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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